

Important Tools

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Table of Contents

1. Introduction	1
2. Topology	1
2.1 Fixed point theorems	1
2.2 Jordan Curve Theorem	2
2.3 Sard's Theorem	3
3. Algebraic Topology	5
3.1 Homotopy Preliminaries	5
3.2 CW Complexes	6
4. Analysis Tools	7
4.1 Urysohn's Lemma	7
4.2 Tietze Extension Theorem	7
4.3 Arzelá-Ascoli Theorem	8
4.4 Partition of Unity	9
4.5 Stone-Weierstraß Theorem	10
4.6 Measure Theory	11
4.7 Riesz-Markov-Kakutani Theorem	14
5. Functional Analysis	16
5.1 C*-algebras	16
5.2 Commutative Gelfand-Naimark Theorem	18
5.3 Continuous functional calculus	19
5.4 Concrete Gelfand-Naimark Theorem	19
5.5 Categorical Perspective	21
6. Homotopy in C*-algebras	22

§1. Introduction

This work is a compilation of useful, but technical theory in topology I would like to remember.

§2. Topology

2.1. Fixed point theorems

Theorem 2.1 (Brouwer's fixed point). Any continuous map $f : D \rightarrow D$ fixes a point for $D = \overline{B_1(0)} \subseteq \mathbb{R}^{n \geq 2}$.

Proof. If not, then $f(x) \neq x$ for all $x \in D$. We can define $g(x)$ as where the ray starting at $f(x)$ going to x intersects ∂D . In particular, $g|_{\partial D} = \text{id}_{\partial D}$. Thus, $S^{n-1} \xrightarrow{i} D \xrightarrow{h} S^{n-1}$ is the identity so $h_* \circ i_* = (h \circ i)_* = \text{id}_{\pi_1(S^{n-1})}$ but $i_* = 0$ since $\pi_1(S^{n-1})$ for $n \geq 2$ – contradiction. $\square$

To generalize this, we need a lemma.

Lemma 2.2. Any convex compact subset A of a real vector space X with nonempty interior is homeomorphic to a ball.

Proof. We assume $0 \in A$ by replacing A with the shifted set $A - a$ for some $a \in A^\circ$. Put $p(x) = \inf\{t > 0 : t^{-1}x \in A\}$. A is compact so bounded, i.e. in $B_R(0)$ for some $R > 0$. Thus $p(x) = \inf\{t > 0 : \|x\|/R \leq |t|\} \leq \|x\|/R < \infty$. Of course p is positive-definite and $\mathbb{R}_{\geq 0}$ -linear. For any x and y , $x/p(x)$ and $y/p(y) \in A$ so $\frac{p(x)}{p(x)+p(y)} \frac{x}{p(x)} + \frac{p(y)}{p(x)+p(y)} \frac{y}{p(y)} = \frac{x+y}{p(x)+p(y)}$ is in A by convexity; thus, $p(x+y) = \inf\{t > 0 : t^{-1}(x+y) \in A\} \leq p(x) + p(y)$ so p is a seminorm.

Therefore, $x \mapsto x \frac{p(x)}{\|x\|}$ provides a homeomorphism to the unit ball for $\|\cdot\|$. $\square$

Theorem 2.3 (Schauder's fixed point). Any continuous map $f : K \rightarrow K$ fixes a point for compact, convex K in a real Banach space X .

Proof. Fix $\epsilon > 0$. By compactness, we can cover K with $B_\epsilon(x_i)$ for $i = 1, \dots, n$. Define continuous functions

$$g_i(x) = \max(0, \epsilon - \|x - x_i\|) \quad \text{and} \quad g(x) = \frac{\sum g_i(x)x_i}{\sum g_i(x)}.$$

Notice the denominator of g never vanishes since these balls cover K so no point is more than ϵ from some x_i at any time. Notice g maps into the convex hull $K' = \{\sum a_i x_i : \sum a_i = 1, a_i \in [0, 1]\}$ of K (where $a_i(x) = g_i(x)/(\sum g_j(x))$).

Since K' is a closed subset of K , it is compact. In particular, it is a compact convex subset of a finite-dimensional real vector space. By 2.2, we can apply 2.1 to $g \circ f|_{K'} : K' \rightarrow K'$ so that $g \circ f(z) = z$ for some $z \in K'$.

Observe $\|g(x) - x\| = \frac{\|\sum g_i(x)(x_i - x)\|}{\|\sum g_i(x)\|} = \frac{\sum g_i(x)\|x_i - x\|}{\sum g_i(x)} \leq \epsilon$ since either $g_i(x) = 0$ or $\|x_i - x\| \leq \epsilon$. Hence, $\|f(z) - z\| = \|f(z) - g(f(z))\| \leq \epsilon$.

Taking $\epsilon = \frac{1}{n}$, we get $z_n \in K$ so $\|f(z_n) - z_n\| \leq 1/n$. By compactness of K , we have some $z_{n_k} \rightarrow z$ so $f(z_{n_k}) \rightarrow z$ but $f(z_{n_k}) \rightarrow f(z)$ by continuity so $f(z) = z$. $\square$

2.2. Jordan Curve Theorem

Definition 2.4 (Jordan Curve). A Jordan curve is an injective continuous map $\gamma : S^1 \rightarrow \mathbb{R}^2$.

There are many proofs of the following deceptively obvious theorem.

Theorem 2.5 (Jordan Curve). $\mathbb{R}^2 - \gamma$ has two components, each of which has boundary γ .

Remark 2.6. Jordan's proof resolved the easy case when γ is polygonal and then carefully approximated Jordan curves by polygons.

We give a short topological proof via Brouwer's fixed point theorem 2.1 per Ryuji Maehara.

Remark 2.7. First observe $\mathbb{R}^2 - \gamma$ has exactly one unbounded component. Namely, γ is compact so bounded, say by $B_R(0)$ so $\mathbb{R}^2 - B_R(0)$ is contained in exactly *the* unbounded component of $\mathbb{R}^2 - \gamma$.

Also, each component of $\mathbb{R}^2 - \gamma$ is open. In particular, any point in a component has a neighborhood in the component disjoint from γ which is path-connected, i.e. each connected open component is locally path-connected, so connected.

Lemma 2.8. If $\mathbb{R}^2 - \gamma$ is disconnected, then each component has γ as its boundary.

Proof. Pick a component A of $\mathbb{R}^2 - \gamma$. Then $\partial A = \overline{A} \cap (\mathbb{R}^2 - A) \subseteq \gamma$. Say $\partial A \neq \gamma$. Thus, ∂A is contained in an arc (i.e. a copy of $[0, 1]$ rather than S^1) σ in γ .

Pick a point p in a bounded component of $\mathbb{R}^2 - \gamma$ — which exists by our remark (and $p \in A$ is A is bounded). Pick some disc D centered at p large enough that ∂D is in the unbounded component of $\mathbb{R}^2 - \gamma$.

Since σ is a copy of $[0, 1]$, we may apply Tietze's extension theorem 4.7 to $\text{id}_\sigma : \sigma \rightarrow \sigma$ for $\sigma \subseteq D$ to get a continuous map $f : D \rightarrow \sigma$. If A is bounded, put $g(x \in \overline{A}) = f(x)$ and $g(x \in A^c) = x$. Otherwise, put $g(x \in \overline{A}) = x$ and $g(x \in A^c) = f(x)$. Note g maps into $D - \{p\}$. Since the intersection of these two closed sets lie in ∂A — on which $f|_{\partial A} = \text{id}_A - g$ is well-defined and continuous by gluing lemma.

Set $\pi : D - \{p\} \rightarrow \partial D$ the obvious projection map and $\nu : \partial D \rightarrow \partial D$ the antipodal map. So $\nu \circ \pi \circ g : D \rightarrow D - \{p\} \rightarrow \partial D \rightarrow \partial D \subsetneq D$ should have a fixed point by 2.1 but it does not – contradiction. $\square$

Lemma 2.9. Any two paths $\eta = (\eta_1, \eta_2), \nu = (\nu_1, \nu_2) : [-1, 1] \rightarrow [a, b] \times [c, d]$ with $\eta_1(-1) = a, \eta_1(1) = b$ and $\nu_2(-1) = c, \nu_2(1) = d$ will meet (possibly at different times).

Proof. Say $\eta(s) \neq \nu(t)$ for all $(s, t) \in [-1, 1]^2$. So $N(s, t) := \max_{i=1,2} |\eta_i(s) - \nu_i(t)| > 0$. Put $f : [-1, 1]^2 \rightarrow [-1, 1]^2$ via $f(s, t) = N(s, t)^{-1}(\nu_1(t) - \eta_1(s), \eta_2(s) - \nu_2(t))$. Note $f \subseteq \partial[-1, 1]^2$. So f has no fixed points because if it did, then either s or t is ± 1 – but the lemma assumption forces the sign of $f(s, t)$ to differ.

Since $[-1, 1]^2$ is a disc, 2.1 leads to a contradiction. So η and ν meet. $\square$

Proof. (Jordan Curve Theorem). By the above remark and first lemma, it suffices to show $\mathbb{R}^2 - \gamma$ has exactly one bounded component.

γ is compact so pick points $a, b \in \gamma$ such that $\|a - b\|$ is greatest. By rotating γ so $a - b$ is horizontal and scaling, we may assume $a = (-1, 0)$ and $b = (1, 0)$. So $[-1, 1] \times [-2, 2]$ contains γ with boundary that meets γ only at a, b .

The straight-line path from $(0, -2) \rightarrow (0, 2)$ will meet γ by applying the previous lemma to the restriction of γ on one of its paths from $a \rightarrow b$. Say ℓ is the y -maximum of $\gamma \cap \{0\} \times [-2, 2]$. Call the half of γ containing ℓ γ^+ and the other path γ^- . Say m is the y -minimum of $\gamma^+ \cap \{0\} \times [-2, 2]$ (where $\ell = m$ is ok). So the straight-line path $m \rightarrow (0, -2)$ meets γ^- by applying the last lemma to γ^- and $m \rightarrow (0, -2)$ with $\ell \rightarrow (0, 2)$ with the portion of γ^+ from $m \rightarrow \ell$ (where these latter two cannot meet γ^- by construction). Say p, q are the y -max and y -min of γ^- and $m \rightarrow (0, -2)$.

Put z_0 the midpoint of p and m and A the component of $\mathbb{R}^2 - \gamma$ containing z_0 .

We show A is bounded. If not, then A being path-connected (per the above remark) would provide a path α from z_0 to outside $[-1, 1] \times [-2, 2]$. Say w was the first point on the boundary of $[-1, 1] \times [-2, 2]$ that α hit. There is a path β along the boundary from w to either $(0, \pm 2)$ without crossing b . Without loss of generality, if boundary path is to $(0, -2)$, then $(0, 2) \rightarrow \ell \xrightarrow{\gamma^+} m \rightarrow z_0 \xrightarrow{\alpha} w \xrightarrow{\beta} (0, -2)$ must meet γ^- by applying the second lemma. By construction of each component, this is only possible if α meets γ^- . So no path from A to outside the rectangle stays in A . Because A is path-connected, this means A is bounded.

If B was a second bounded component, it certainly would be in $[-1, 1] \times [-2, 2]$ as the complement of this rectangle – one unbounded component bounds γ . But $(0, 2) \rightarrow \ell \xrightarrow{\gamma^+} m \rightarrow p \xrightarrow{\gamma^-} q \rightarrow (0, -2)$ does not meet B or a or b . In particular, there are balls around a and b which do not meet this composite path. Yet the first lemma implies $\bar{B}$ is γ , so these balls intersect B , say at a' and b' . B is path-connected so there is a path ρ from $a' \rightarrow b'$ in B such that $a \rightarrow a' \xrightarrow{\rho} b' \rightarrow b$ must meet our composite path by the second lemma. But none of the three segments can do so by construction – contradiction. Thus, there is exactly one boundary component of $\mathbb{R}^2 - \gamma$. $\square$

2.3. Sard's Theorem

We assume basic knowledge of manifolds.

Definition 2.10 (Critical point/value). x is a **critical point** of a C^k map $f : M^m \rightarrow N^n$ if df_x is not full-rank, i.e. if $df_x(T_x M) \neq T_{f(x)} N$. In this case $f(x)$ is a **critical value**.

Definition 2.11 (Measure zero). A set $S \subseteq N^n$ is **measure zero** if it has zero measure (see: 4.6) in $\mathbb{R}^n$ (equipped with the Lebesgue measure μ) under every chart.

Theorem 2.12 (Sard). The set of critical values of a C^k map $f : M^m \rightarrow N^n$ is measure zero.

Remark 2.13. We can cover M, N by a countable number of charts so, for the proof, it suffices to consider a C^k map $f : U \rightarrow \mathbb{R}^n$ for an open set $U \subseteq \mathbb{R}^m$.

To continue, we need the following definition and lemmas.

Definition 2.14 (C, C_i) . Put C_i the set of all x such that all partial derivatives of order at most i of f vanish at x . Observe $C_1 = \{x : df_x = 0\}$. Say C is the set of critical values so

$$C \supseteq C_1 \supseteq C_2 \supseteq \cdots.$$

Lemma 2.15 (1). $f(C - C_i)$ has measure zero.

Proof. Fix $x \in C - C_1$. If we can find open sets V_x so $f(V_x \cap C)$ is measure zero, then we win since we can find a countable number of open sets covering $C - C_1$, i.e.

$$\mu(f(C - C_1)) \leq \sum_i \mu(f([C - C_1] \cap V_{x_i})) \leq \sum_i \mu(f(C \cap V_{x_i})) = 0.$$

We prove $f((C - C_i) \cap V)$ has measure zero by induction on n . For $n = 1$, $C = C_1$ so $f(C - C_1)$ is measure zero.

Now suppose we've shown $f((C - C_1) \cap V)$ has measure zero for $n' = 1, \dots, n - 1$. Since $x \notin C_1$, some $\partial_i f(x) \neq 0$. Set $h : U \rightarrow \mathbb{R}^m$ by $h(x) = (f_1(x), x_2, \dots, x_m)$. dh_x is non-singular at x so rank-theorem implies there is a diffeomorphism of some neighborhood V of x with a neighborhood V' of $h(x)$. Thus $g = f \circ h^{-1} : V' \rightarrow \mathbb{R}^n$ and $f|_V$ have the same critical values.

Note $dg = \begin{pmatrix} 1 & 0 \\ ? & \partial_{j>1} g^{i>1} \end{pmatrix}$ on V' so g has a critical point if and only if $g^{>i}$ has a critical point.

Therefore, if we let V^t denote the set of critical values for which g^t has measure zero, then (this set is closed so measurable and Tonelli's theorem gives):

$$\mu(V) = \int_{\mathbb{R}^n} \chi_V d\mu = \int_{\mathbb{R}} \int_{\mathbb{R}^{n-1}} \chi_{V^t} d\mu_{n-1} dt = \int_{\mathbb{R}} 0 dt = 0.$$

□

Lemma 2.16 (2). $f(C_k - C_{k+1})$ is measure zero for $k \geq 1$.

Proof. For each $x \in C_k - C_{k+1}$, some $k + 1$ -order partial of f does not vanish at f so there is a k th partial ρ for which, say, $\partial_1 \rho(x) \neq 0$. Thus, we may define $h : U \rightarrow \mathbb{R}^m$ via $h(x) = (\rho(x), x_2, \dots, x_m)$ as before and get some V, V', g as before so that g and $f|_V$ have the same critical values but clearly all the critical values of g come from $0 \times \mathbb{R}^{m-1}$.

Thus the critical points of $g|_{0 \times \mathbb{R}^{m-1}}$ matches the critical points of g of type C_k – so we are done by induction. □

Lemma 2.17 (3). For $k > m/n - 1$, $f(C_k)$ is measure zero.

Proof. Consider a cube X of side-length δ in U . U is covered by countably many copies of X , so it suffices to show $f(C_k \cap X)$ has measure zero.

For an as-yet unknown N , subdivide X into N^m cubes of side-length δ/N . By the compactness of X , some constant $A > 0$ bounds the absolute values of all the $(k + 1)$ -derivatives of f in X . So on the intersection of C_k and any of these smaller cube X' :

$$|f(x + h) - f(x)| < A|h|^{k+1} \leq A|\delta/N|^{k+1}.$$

That is, $f(C_k \cap X_k)$ fits into a ball of radius $A|\delta/N|^{k+1}$ so a cube of side-length $A2^n|\delta/N|^{n(k+1)}$. Since there are N^m such cubes, we can upper bound this volume by $2^n \delta^{n(k+1)} N^{m-n(k+1)}$. Yet $n(k+1) > m$ by assumption so we can make N as large as possible shrinking the volume. Thus, $f(C_k \cap X)$ has measure zero. □

Proof. (Sard's Theorem). Simply apply the last remark and 3 lemmas to see that $f(C - C_1) + \sum_1^{m/n-1} f(C_i - C_{i+1}) + \sum_{m/n}^\infty f(C_i)$ has measure zero. □

§3. Algebraic Topology

3.1. Homotopy Preliminaries

We start with some terminology.

Definition 3.1 (CX). The cone CX is $(X \times I)/(X \times 1)$.

Definition 3.2 (ΣX). The suspension ΣX is $CX/(X \times 0)$.

Remark 3.3. We can change these to base-point spaces $C(X, x_0)$ and $\Sigma(X, x_0)$ by further quotienting via $x_0 \times I$ such that all possible basepoints collapse to a single point.

Definition 3.4 ($C(X, Y)$). We can give the continuous maps $C(X, Y)$ of X into Y has the compact-open topology, i.e. a base of open sets $U(K, O)$ (for K a compact subset of X and an open set O of Y) containing maps $f \in C(X, Y)$ so that $f(K) \subseteq O$.

Definition 3.5 ($\Omega(X, x_0)$). We denote loop space as $\Omega(X, x_0) = C((S^1, 1), (X, x_0)) \subseteq C(S^1, X)$.

Proposition 3.6. For base-point spaces $(X, x_0), (Y, y_0)$, there is a natural (over X, Y) homeomorphism

$$C(\Sigma X, Y) = C(X, \Omega Y).$$

Proof. Consider $f \in C(\Sigma X, Y)$, i.e. $f : X \times I \rightarrow Y$ with $f(x, 0) = f(x_0, t) = f(x, 1)$. In particular, we can view f as a map $x \mapsto (f_x : (S^1, 1) \rightarrow (Y, y_0))$. This is clearly a homeomorphism.

Natural simply means that (base-point preserving maps) $X' \rightarrow X$ and $Y' \rightarrow Y$ produce a commutative diagram:

$$\begin{array}{ccc} C(\Sigma X, Y) & \longrightarrow & C(\Sigma X', Y) \\ \downarrow & & \downarrow \\ C(\Sigma X, Y') & \longrightarrow & C(\Sigma X', Y') \end{array}$$

□

Definition 3.7 (Homotopy). $f, g : X \rightarrow Y$ are **homotopic** if there is some continuous map $H : X \times I \rightarrow Y$ so that $H_0(x) = f(x)$ and $H_1(x) = g(x)$. This is an equivalence relation.

Definition 3.8 ($\pi(X, Y)$). $\pi(X, Y)$ (or $[X, Y]$) denotes the homotopy classes of $C(X, Y)$.

$\pi_b(X, Y)$ denotes the base-point version.

Remark 3.9. Maps $f : X' \rightarrow X$ and $g : Y \rightarrow Y'$ produce maps $f_* : \pi(X, Y) \rightarrow \pi(X', Y)$ and $g_* : \pi(X, Y) \rightarrow \pi(X, Y')$.

Remark 3.10. For $f : X \rightarrow X'$ and $g : X' \rightarrow X''$, $g_* \circ f_* : \pi(X, Y) \rightarrow \pi(X'', Y)$ matches $(g \circ f)_*$.

Definition 3.11 (Homotopy equivalent). X, Y are homotopy equivalent if there are maps $f : X \rightarrow Y, g : Y \rightarrow X$ such that $g \circ f \sim \text{id}_X$ and $f \circ g \sim \text{id}_Y$.

Definition 3.12 (Contractible). A space X is contractible if X is homotopy equivalent to a point, i.e. if and only if $\text{id}_X \sim x_0$.

Definition 3.13 (((Strong) deformation) retract). A map $r : (X, A) \rightarrow (X, A)$ is a retract if $r(X) = A$ and $r|_A = \text{id}_A$ for a subspace $A \subseteq X$. r is a deformation if $r \stackrel{H}{\sim} \text{id}_X$. r is a strong reformation if there is a homotopy $H_t|_A = \text{id}_A$ for all $t \in [0, 1]$.

Remark 3.14. A point is a deformation retract if and only if X is contractible.

Definition 3.15 (Contractible in X). A subspace A is contractible in X if the inclusion $A \rightarrow X$ is homotopic to a constant map $A \rightarrow X$.

Definition 3.16 ($\text{cat} X$). The Lusternik-Schnirelmann category of X , $\text{cat} X$ is the minimal $n \in [1, \infty]$ such that X can be covered by n open subsets contractible in X .

3.2. CW Complexes

Definition 3.17 (CW complex). A CW complex is a Hausdorff space $X = \bigcup_{q=0}^{\infty} \bigcup_{i \in I_q} e_i^q$ where e_i^q are pairwise disjoint cells so e_i^q is a copy of the open q -ball B^q attached via a map $f_i^q : \overline{B^q} \rightarrow X$ so $\partial B^q = S^{q-1}$ maps into $\bigcup_{q'=0}^{q-1} \bigcup_{i \in I_{q'}} e_i^{q'}$ such that: (C) $\partial e_i^q = f_i^q(S^{q-1})$ is contained in a finite union of cells and (W) a set $A \subseteq X$ is closed if and only if $A \cap \overline{e_i^q}$ is closed, i.e. if $(f_i^q)^{-1}(A)$ is closed in $\overline{B^q}$ for all i, q .

Definition 3.18 (Subcomplex). Y is a subcomplex of X if it is a closed subset of X made of whole cells. (X, Y) is a CW pair.

Definition 3.19 (n th skeleton). $\text{sk}_n X = X^n = \bigcup_{q=0}^n \bigcup_{i \in I_q} e_i^q$ is the n th-skeleton of a CW complex X .

Definition 3.20 (Cellular). A map $f : X \rightarrow Y$ is cellular if $f(X^n) \subseteq Y^n$ for all n .

Remark 3.21. A CW complex X is path-connected if and only if X^1 is path-connected if and only if it is connected.

Definition 3.22 (Borsuk pair). (X, A) is a Borsuk pair if for every space $F_0 : X \rightarrow Y$ and homotopy $f_t : A \rightarrow Y$ so $F_0|_A = f_0$, that there is a homotopy $F_t : X \rightarrow Y$ so $F_0 = F$ and $F_t|_A = f_t$.

Theorem 3.23 (Borsuk). CW pairs are Borsuk pairs.

Proof. We can consider f_t as some $f : A \times I \rightarrow Y$. We will build $F : X \times I \rightarrow Y$ by induction.

Put $F(x \in A, t) = f(x, t)$ and $F(x \text{ 0-dim cell } \notin A, t) = x$.

Now say F is defined on $(A \cup X^n) \times I$ and agrees with f and F_0 . Fix some cell $e^{n+1} \subseteq X - A$ (recall A is a subcomplex so made of whole cells so this is an okay assumption). Note F is defined on $\partial e^{n+1} \times I \cong S^n \times I$ and F_0 is defined on $\overline{B^{n+1}} \times 0$. In particular, this is a hollow cylinder with exactly one base $\overline{B^{n+1}}$. Project any point in the solid cylinder to $(S^n \times I) \cup (\overline{B^{n+1}} \times I)$ along a ray starting at $(0, 3/2)$ – note this is the identity on $(S^n \times I) \cup (\overline{B^{n+1}} \times I)$. This lets us define a map $\overline{B^{n+1}} \times I \rightarrow Y$, i.e. $F|_{e^{n+1} \times I}$ which agrees with what has been defined already.

Doing this for all $n+1$ -cells finishes the inductive step. F will be continuous for complexes with cells of arbitrarily large dimension by (W). $\square$

Corollary 3.24. If A is contractible for (X, A) a CW pair, then $\pi : X \rightarrow X/A$ is a homotopy equivalence.

Proof. A is contractible so $\text{id}_A \stackrel{f_t}{\sim} a$ for some $a \in A$. $F_0 = \text{id}_X$ and $F_0|_A = f_0$ so, since (X, A) is a Borsuk pair, there is a homotopy $F : X \times I \rightarrow X$ for which $F_t|_A = f_t$ and $F_0 = \text{id}_X$.

Notably, $F_1(A) = a$. so $F_1 : X \rightarrow X$ factors as $F_1 : X \xrightarrow{\pi} X/A \xrightarrow{q} X$ for some q . So $q \circ \pi = F_1 \sim F_0 = \text{id}_X$.

On the other hand, $f_t(A) \subseteq A$ so $F_t(A) \subseteq A$ so F_t descends to a homotopy $F_t/A : X/A \rightarrow X/A$ such that $\pi \circ F_t = F_t/A \circ \pi$ so $\text{id}_{X/A} \circ \pi = \pi \circ F_0 \sim \pi \circ F_1 = \pi \circ (q \circ \pi) = (\pi \circ q) \circ \pi$. $\square$

Corollary 3.25. If (X, A) is a CW pair, then $X/A \sim X \cup CA$.

Proof. Observe CA is contractible (to its tip) and a CW subcomplex. Applying the above corollary to $X \cup CA \rightarrow X \cup CA/CA = X/A$ finishes. $\square$

Remark 3.26. I learned the following proof from Professor Alexander Givental.

Theorem 3.27 (Cellular Approximation). Any continuous map between CW complexes is homotopic to a cellular map.

We need a lemma first.

Lemma 3.28 (Free-point). Given an open set $U \subseteq \mathbb{R}^n$ and $f : U \rightarrow B^{m>n}$ the open m -ball such that $K_0 = f^{-1}d^m \subseteq U$ is compact for a closed ball $d^m \subseteq B^m$, there is a continuous map $F : U \rightarrow B^q$ that agrees with f in the complement of $U - K$ such that the image of F does not cover all of d^m .

Proof. Think of f as a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$ (the blown up interior of B^m). And assume $f(x) = y \in d^m$ for some y (since if f does not meet d^m , we are already done).

Take an open ball b_2 around y in d^m . Pick some $\epsilon > 0$ so that the closure of a ball b_1 of radius $\text{rad}(b_2) + \epsilon$ centered at y also stays in d^m . Say $K_1 = f^{-1}b_1$.

By Stone-Weierstrass 4.24, there is a polynomial p so $\|f - p\|_{K_0} < \epsilon$. Also use 4.18 to take a bump function ρ with $\text{supp} \rho \subseteq d^m$ that is 1 and on $\overline{b_1}$. Clearly $\tilde{f} = f + (p - f)\rho$ satisfies $\|f - \tilde{f}\| < \epsilon$. Notably, $\tilde{f}(K_0 - K_1) \cap b_2$ is empty so $\tilde{f}|_{K_1} = p$ so $\tilde{f}(K_1) \cap b_2$ is zero-measure by Sard's theorem. Take the homotopy $f_t = (1 - t)f + t\tilde{f}$. $\square$

§4. Analysis Tools

4.1. Urysohn's Lemma

Definition 4.1 (Normal). A topological space (X, τ) is **normal** if for any two disjoint closed sets V, V' , there are disjoint open sets U, U' with $V \subseteq U, V' \subseteq U'$.

Lemma 4.2. If (X, τ) is normal, then for a closed set V contained in an open set U , there is another open set W containing V so $V \subseteq W \subseteq \overline{W} \subseteq U$.

Proof. By (X, τ) normal, we get disjoint open sets O_V, O_{U^c} containing closed sets V, U^c . In particular, $V \subseteq O_V \subseteq \overline{O_V} \subseteq O_{U^c}^c \subseteq U$. $\square$

Theorem 4.3 (Urysohn). If (X, τ) is normal, then for disjoint closed subsets V_0, V_1 , there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f|_{V_0} = 0, f|_{V_1} = 1$.

Proof. Let $U_1 = V_1^c$. By 4.2 and $V_0 \subseteq U_1$, we get $V_0 \subseteq U_{1/2} \subseteq \overline{U_{1/2}} \subseteq U_1$. Inductively, we can repeat this with $V_0 \subseteq U_{1/2}$ and $\overline{U_{1/2}} \subseteq U_1$, to get a chain of open sets $U_{k/2^n}$ for $k/2^n \in (0, 1]$. Define:

$$f(x) = \begin{cases} \inf\{\frac{k}{2^n} : x \in U_{k/2^n}\} & x \notin V_1, \\ 1 & x \in V_1. \end{cases}$$

Clearly, $f(V_0) = 0$. We show f is continuous by checking a sub-base $[0, a)$ and $(b, 1]$; $f^{-1}[0, a) = \bigcup_{a \leq k/2^n} U_{k/2^n}$ and $f^{-1}(b, 1] = \bigcup_{b < k/2^n} \overline{U_{k/2^n}}$ are open. $\square$

Corollary 4.4. A space (X, τ) is normal if and only if there exists an Urysohn function separating any 2 disjoint closed subsets of X .

Remark 4.5. Compact Hausdorff spaces are normal.

4.2. Tietze Extension Theorem

Lemma 4.6. Given a closed subset A of a normal topological space (X, τ) and a continuous function $f : A \rightarrow [0, r > 0]$, there is a continuous map $g : X \rightarrow [0, r/3]$ with $0 \leq f - g|_A \leq 2r/3$.

Proof. Consider closed subsets $B = f^{-1}[0, r/3]$ and $C = f^{-1}[2r/3, r]$. Urysohn's lemma 4.3 implies the existence of $\tilde{g} : X \rightarrow [0, 1]$ continuous with $\tilde{g}|_B = 0, \tilde{g}|_C = 1$. $g(x) = \frac{r}{3}\tilde{g}(x)$ suffices. $\square$

Theorem 4.7 (Tietze extension). Given a closed subset A of a normal topological space (X, τ) , any continuous function $f : A \rightarrow \mathbb{R}$ can be extended to a continuous map $\tilde{f} : X \rightarrow \mathbb{R}$ with $\text{im} \tilde{f} \subseteq [a, b]$ if $\text{im} f \subseteq [a, b]$.

Proof. We can assume $\text{im} f \subseteq [0, 1]$ by post-composing with a homeomorphism $\mathbb{R} \cong (0, 1)$.

If $r = 1$, 4.6 implies the existence of $g_1 : X \rightarrow [0, 1/3]$ with $0 \leq f - g_1 \leq 2/3$. Applying the lemma to $f - g_1|_A$ produces some $g_2 : X \rightarrow [0, (2/3)^2]$ with $0 \leq (f - g_1|_A) - g_2|_A \leq (2/3)^2$. By induction, $(g_n)_n$ forms a Cauchy sequence with respect to the supremum norm. In particular, $(\sum_1^n g_i)_n$ is Cauchy and $[0, 1]$ is complete so this sequence will converge to some continuous $g : X \rightarrow \mathbb{R}$ with $\text{img} \subseteq [0, 1]$ for which $g|_A \equiv f$. $\square$

4.3. Arzelá-Ascoli Theorem

Definition 4.8 (Totally bounded). A subset A of a metric space (X, d) is **totally bounded** if for any $\epsilon > 0$, there is a finite number of points $(x_i)_1^n \subseteq X$ for which $A \subseteq \cup_1^n B_\epsilon(x_i)$.

Lemma 4.9. A subset A of a totally bounded space (X, d) is totally bounded.

Proof. Fix $\epsilon > 0$. We get $(x_i)_1^n \subseteq X$ for which $X \subseteq \cup_1^n B_{\epsilon/2}(x_i)$. For each ball that intersects A , pick a point y_i in said intersection. Thus, $x \in A$ implies $x \in B_{\epsilon/2}(x_i)$ for some i so $d(x, y_i) \leq d(x, x_i) + d(x_i, y_i) < \epsilon/2 + \epsilon/2 = \epsilon$, i.e. $A \subseteq \cup B_\epsilon(y_i)$. $\square$

Theorem 4.10. A metric space (X, d) is compact if and only if it is complete and totally bounded.

Proof. $\implies$: Say X is compact. If (x_n) is Cauchy, by the compactness of X , (x_n) has a convergent subsequence with a unique limit point, i.e. X is complete. Totally bounded is an immediate consequence of compactness.

$\impliedby$: Say X is not compact, but is complete and totally bounded. Pick an open cover $\mathcal{U}$ of X with no finite subcover. Plus, for any ϵ , a finite number of ϵ -balls cover X . All of the balls cannot be covered by a finite subcover of $\mathcal{U}$ by assumption, so we find x_1 with $B_{1/2}(x_1)$ not compact, but totally bounded by 4.9. Applying our observation again with balls of radius $1/4, 1/8, \dots$, we get a Cauchy sequence (x_n) . X is complete so this sequence converges to some x . But a ball centered at x would cover all but finitely many x_n , breaking non-compactness - contradiction. $\square$

Definition 4.11 (Equicontinuous). A family of continuous functions $\mathcal{F}$ from a topological space (X, τ) to a metric space (X, d) is **equicontinuous at some $x \in X$** if for all ϵ , there is some open neighborhood U of x for which $d(f(x), f(U)) < \epsilon$ on $\mathcal{F}$. $\mathcal{F}$ is **equicontinuous** if it is equicontinuous everywhere.

Theorem 4.12 (Arzelá-Ascoli). Equip the (bounded) continuous functions $C(X, M)$ from a compact topological space into a metric space with the supremum metric d_u . Then any equicontinuous and pointwise totally bounded family $\mathcal{F} \subseteq C(X, M)$ is totally bounded.

Proof. Fix $\epsilon > 0$. By equicontinuity, for every $x \in X$, we get some U_x on which $d(f(x), f(U)) < \epsilon/4$ for all $f \in \mathcal{F}$. By compactness of X , we may reduce to a finite number n of $U_i := U_{x_i}$. $\mathcal{F}(x_i)$ is totally bounded so we get some finite set S_i of functions for which $\mathcal{F}(x_i) \subseteq \cup_{f \in S_i} B_{\epsilon/4}(f(x_i))$. $S = \cup_1^n S_i$ is also finite.

Fix $\psi \in \prod_1^n S$ and let

$$\mathcal{F}_\psi = \{f \in \mathcal{F} : d(f(x_i), \psi_i(x_i)) < \epsilon/4 \text{ for } 1 \leq i \leq n\}.$$

By the first paragraph, clearly, $\mathcal{F} \subseteq \bigcup_{\psi \in \prod_1^n S} \mathcal{F}_\psi$. It suffices to show $\mathcal{F}_\psi$ has diameter ϵ (and thus radius $\leq \epsilon$). That is, pick $f, g \in \mathcal{F}_\psi$. For any $x \in X$, $x \in B_{\epsilon/4}(x_i)$ for some i so

$$\begin{aligned} d_u(f, g) &= \sup_{x \in X} d(f(x), g(x)) \\ &\leq \sup_{x \in X} (d(f(x), f(x_i)) + d(f(x_i), \psi_i(x_i)) + d(\psi_i(x_i), g(x_i)) + d(g(x_i), g(y))) \leq \epsilon. \end{aligned}$$

$\square$

Lemma 4.13. Given a set X and a complete metric space (M, d) , the space of bounded functions $B(X, M)$ is complete under the supremum metric d_u .

Notice there is no idea of continuity in the above lemma since X is not topological.

Proof. Fix $(f_n) \subseteq B(X, M)$ Cauchy under d_u . Since, $d(f_n(x), f_m(x)) \leq d_u(f_n, f_m)$, $(f_n(x))$ is Cauchy in M for fixed x so will converge to some x . Define $f(x) := \lim f_n(x)$.

We show f is bounded. Because $d|_{f_m(x)} : X \rightarrow \mathbb{R}$ is continuous,

$$d_u(f, f_m) = \sup_{x \in X} d(\lim_{n \rightarrow \infty} f_n(x), f_m(x)) = \sup_{x \in X} \lim_{n \rightarrow \infty} d(f_n(x), f_m(x)).$$

Since (f_n) is Cauchy under d_u , there exists m such that $d(f_{n>m}(x), f_m(x)) < \epsilon$ for any $\epsilon > 0$. In particular, this holds as $n \rightarrow \infty$ so $d_u(f, f_m) \leq \sup_{x \in X} \epsilon = \epsilon$.

Hence, if $\text{im} f_m$ is covered by a ball of radius r , then $\text{im} f$ is covered by a ball of radius $r + \epsilon$. $\square$

In particular, Arzelà-Ascoli matters for the following reason.

Corollary 4.14. If X is a compact topological space and (M, d) is a complete metric space, then a subset $\mathcal{F} \subseteq C(X, M) \subseteq B(X, M)$ is compact if and only if it is equicontinuous, closed, and bounded.

Proof. $\implies$: Say $\mathcal{F}$ is compact. $\mathcal{F}$ is closed since it is a compact subset of a Hausdorff (because of the metric d_u) space $C(X, M)$. $\mathcal{F}$ is also bounded since it is totally bounded by 4.10.

Lastly, suppose $\mathcal{F}$ is not equicontinuous. That is, there is some $x \in X$ and $\epsilon > 0$ for which, given any neighborhood U of x , there is some f_U for which $d(f_U(x), f_U(U))$ exceeds ϵ . The net (f_U) has a cluster point f in $\mathcal{F}$ by compactness. Let $V = f^{-1}B_{\epsilon/3}(f(x))$. We can find f_U with $d_u(f, f_U) < \epsilon/3$, i.e. for any $y \in V \cap U$,

$$d(f_U(x), f_U(y)) \leq d(f_U(x), f(x)) + d(f(x), f(y)) + d(f(y), f_U(y)) < \epsilon \text{ as desired.}$$

$\impliedby$: Say $\mathcal{F}$ is equicontinuous, closed, and bounded. Hence, 4.12 applies and $\mathcal{F}$ is totally bounded. $B(X, M)$ is complete by 4.13. Since $\mathcal{F}$ is a closed subset of this complete space, it is also complete. Hence, 4.10 applies and $\mathcal{F}$ is compact. $\square$

4.4. Partition of Unity

We reference one more tool.

Definition 4.15 (Locally compact). A topological space X is locally compact if every point has a neighborhood with compact closure.

Definition 4.16 (Support). If X is locally compact Hausdorff and $(V, \|\cdot\|)$ is a normed k -vector space and $f \in C(X, V)$, the **support** of f is $\overline{f^{-1}(0)}$.

Definition 4.17 ($C_c(X, V)$). $\underline{C_c(X, V)}$ is the k -subalgebra of $C(X, V)$ consisting of functions with compact support.

Proposition 4.18. If K is a compact subset of an open set O in a locally compact Hausdorff space X , then there is a continuous function $f : X \rightarrow \mathbb{R}$ with $f|_K = 1$ and $\text{supp} f \subseteq O$.

Proof. For each $x \in K$, there is an open neighborhood U_x of x with compact closure. $\overline{U_x} \cap O$ is closed and thus compact since it is a subset of $\overline{U_x}$. In particular, $\overline{U_x} \cap O$ is compact Hausdorff so is normal. Notice $\overline{U_x} \cap O - (U_x \cap O)$ is closed in $\overline{U_x} \cap O$ and $\{x\}$ is closed since our space is Hausdorff so, by normality, we can find some open set W_x containing x whose closure is in $U_x \cap O$ (and obviously compact).

We can cover K with a finite number of $W_i = W_{x_i}$ since K is compact. Hence $K \subseteq W = \cup_1^n W_i \subseteq \overline{W} = \cup_1^n \overline{W_i} = \cup_1^n \overline{W_i}$ is compact Hausdorff and contained in O .

Because $\overline{W}$ is normal, 4.3 promises us a function $f : \overline{W} \rightarrow \mathbb{R}$ with $f|_K \equiv 1$ and $f|_{\overline{W}-W} \equiv 0$. Put $\tilde{f} : X \rightarrow \mathbb{R}$ that agrees with f on $\overline{W}$ and vanishes outside $\overline{W}$.

We need only check $\tilde{f}$ is continuous. For any open set $A \subseteq \mathbb{R}$, if A does not contain 0, then $\tilde{f}^{-1}A \subseteq \overline{W}$ in which case it agrees with $f^{-1}A$ so is already open. If A contains 0, then $\tilde{f}^{-1}A = f^{-1}A \cup (X - \overline{W})$ so is also open. $\square$

Proposition 4.19. If $(U_i)_1^n$ is an open cover of a compact subset K of a locally compact Hausdorff space X , then there is partition of unity on K subordinate to U_i of compactly supported functions.

Proof. Just like the first paragraph of the proof of 4.18, we get compact $K_i \subseteq U_i$. By Urysohn's lemma, we get $g_i \in C_c(X, [0, 1])$ with $g_i|_{K_i} \equiv 1$ and $\text{supp } g_i \subseteq U_i$. $\sum_1^n g_i \geq 1$ on K so applying Urysohn's to $K \subseteq (\sum_1^n g_i)^{-1}(0, \infty)$ once more gives some $f \in C_c(X, [0, 1])$ with $f|_K \equiv 1$ and $\text{supp } f \subseteq (\sum_1^n g_i)^{-1}(0, \infty)$. If $g_{n+1} = 1 - f$, then $h_i := \frac{g_i}{\sum_1^{n+1} g_i}$ satisfies $\text{supp } h_i = \text{supp } g_i \subseteq U_i$ and $\sum_1^n h_i = 1 - \frac{g_{n+1}}{\sum_1^{n+1} g_i}$ is 1 on K . $\square$

Remark 4.20. $C_c(X, \mathbb{R})$ is a subalgebra of the complete space of bounded continuous functions $X \rightarrow \mathbb{R}$. But, $C_c(X)$ is not closed; in fact its closure is $C_0(X)$, the functions which vanish at ∞ , i.e. for any $\epsilon > 0$, there is a compact subset outside which $|f| < \epsilon$.

4.5. Stone-Weierstraß Theorem

For this section, X denotes a locally compact Hausdorff space and $C_0(X)$ denotes the real-valued functions vanishing at ∞ .

Lemma 4.21. If $f \in C_0(X)$ is non-negative, polynomials in f can approximate $\sqrt{f}$.

Proof. Fix $\epsilon > 0$. There is a compact set K outside which $|f| \leq \epsilon$. So $f(K) \subseteq [m, M] \subseteq [0, \infty)$. Hence, $\text{im } f \subseteq [0, M + \epsilon]$. Since $\sqrt{y} = \sum_{k=0}^{\infty} \binom{1/2}{k} (y - 1)^k$ on $[1, 2]$ and each term in the summation is non-negative and decreasing, we can truncate the series to get some $p \in \mathbb{R}[x]$ for which $\|\sqrt{\frac{f}{M+\epsilon}} + 1 - p(f)\|_{\infty} < \epsilon$. So there is some $\tilde{p} \in \mathbb{R}[x]$ with $\|f - \tilde{p}\|_{\infty} < \epsilon$. $\square$

Corollary 4.22. If $f \in C_0(X)$, then $|f| = \sqrt{f^2}$ can be approximated by polynomials in f .

Corollary 4.23. If $f, g \in C_0(X)$, then $\max(f, g) = \frac{f+g+|f-g|}{2}$ and $\min(f, g) = \frac{f+g-|f-g|}{2}$ can be approximated by polynomials in f, g .

Theorem 4.24 (Stone-Weierstraß). Say X is a locally compact Hausdorff space and A is a subalgebra of $C_0(X, \mathbb{R})$. Then A is dense in $C_0(X, \mathbb{R})$ under the supremum metric if and only if it separates points and vanishes nowhere.

Proof. $\implies$: Fix $x \neq y$. Points are compact, and thus closed, since X is Hausdorff. By the proof of 4.18, we can produce a continuous function $f \in C_0(X, \mathbb{R})$ with $f|_{\{x\}} \equiv 1$ and $\text{supp } f \subseteq X - \{y\}$. A is dense in $C_0(X, \mathbb{R})$ so there is some $g \in A$ with $\|g - f\| < 1/3$. Hence, $g(x) > 2/3$ and $g(y) < 1/3$. Thus, A separates points and vanishes nowhere (namely, x).

$\impliedby$: It suffices to show $A = C_0(X, \mathbb{R})$ if A is closed. Suppose $h \in C_0(X, \mathbb{R})$ and fix $\epsilon > 0$. Fix x, y . We will produce some $f_{xy} \in A$ that agrees with h at x, y .

We can find some $f \in A$ which does not vanish at x and some $g \in A$ which differs at x, y . It follows that

$$f_{xy}(z) = \left(h(y) - h(x) \frac{f(y)}{f(x)} \right) \frac{g(z) - g(x)}{g(y) - g(x)} + \frac{h(x)}{f(x)} f(z) \in A.$$

Consider the open neighborhood $U_x = \{z \in X : h(z) < f_{xy} + \epsilon\}$ of x . Since $h, f_{xy} \in C_0(X, \mathbb{R})$, i.e. vanish at ∞ , there is a compact set K outside which $h - f_{xy} < \epsilon$. Hence, the closed set $X - U_x \subseteq K$ is actually compact. Fix some $x_0 \in X$ and take a finite sub-cover $X - U_{x_0} \subseteq U_{x_1} \cup \dots \cup U_{x_n}$. Let $f_y = \max(f_{x_1, y}, \dots, f_{x_n, y}) \in A$ so $f_y \geq f_{x_i, y} > h - \epsilon$.

Repeat this process with $h - f_y > -\epsilon$ and take the minimum to produce some $f \in A$ with $\|f - h\|_{\infty} < \epsilon$. Because A is closed and ϵ is arbitrary, $h \in A$. $\square$

4.6. Measure Theory

Definition 4.25 (σ -algebra). $M \subseteq \mathcal{P}(X)$ is a σ -algebra of a non-empty set X if M is a non-empty collection closed under countable unions and complements.

Remark 4.26. Since M is non-empty, it must contain X and $\emptyset$.

Remark 4.27. The intersection of σ -algebras is still a σ -algebra so we can consider the intersection of all σ -algebras containing some subset E of $\mathcal{P}(X)$, i.e. the smallest σ -algebra containing E , i.e. the σ -algebra *generated* by E .

Definition 4.28 (Borel σ -algebra). The Borel σ -algebra of a topological space X is the σ -algebra generated by all the open sets, i.e. the topology, of X . Members of this σ -algebra are called Borel sets.

Definition 4.29 (Measure). A measure on a set X equipped with a σ -algebra M is $\mu : M \rightarrow [0, \infty]$ satisfying $\mu(\emptyset) = 0$ and countable additivity, i.e. if $(E_i)_{i=1}^\infty \subseteq M$ are disjoint then $\mu(\cup E_i) = \sum \mu(E_i)$.

We call (X, M, μ) a measure space.

Definition 4.30 (σ -finite). E is σ -finite for μ if it is the countable union of sets of finite μ -measure.

Many of the nice properties we'd expect of a measure hold.

Proposition 4.31. Say (X, M, μ) is a measure space. Then:

- (a) If $E, F \in M$ and $E \subseteq F$, then $\mu(E) \leq \mu(F)$.
- (b) If $(E_i) \subseteq M$, then $\mu(\cup E_i) \leq \sum \mu(E_i)$.
- (c) If $(E_i) \subseteq M$ and $E_1 \subseteq E_2 \subseteq \dots$, then $\mu(\cup E_i) = \lim \mu(E_i)$.
- (d) If $(E_i) \subseteq M$ and $\mu(E_1) < \infty$ and $E_1 \supseteq E_2 \supseteq \dots$, then $\mu(\cap E_i) = \lim \mu(E_i)$.

Proof. (a) $\mu(E) \leq \mu(E) + \mu(F - E) = \mu(F)$.

(b) Put $F_1 = E_1$ and $F_i = E_i - \cup_{j=1}^{i-1} E_j$. Thus, $\mu(\cup_1^\infty E_i) = \mu(\cup_1^\infty F_i) = \sum_1^\infty \mu(F_i) \leq \sum_1^\infty \mu(E_i)$.

(c) Calling $E_0 = \emptyset$, then $\mu(\cup_1^\infty E_i) = \mu(\cup_1^\infty (E_i - E_{i-1})) = \sum_1^\infty \mu(E_i - E_{i-1}) = \lim \sum_1^n \mu(E_i - E_{i-1}) = \lim \mu(E_n)$.

(d) $\mu(E_1) = \mu(\cap_1^\infty E_i) + \mu(\cup_1^\infty (E_1 \cap E_i^c)) \stackrel{(c)}{=} \mu(\cap_1^\infty E_i) + \lim \mu(E_1 \cap E_i^c) = \mu(\cap_1^\infty E_i) + \lim [\mu(E_1) - \mu(E_i)]$ so subtracting $\mu(E_1)$ from both sides gives $\lim \mu(E_i) = \mu(\cap_1^\infty E_i)$. $\square$

Definition 4.32 (Null set). E is a null set of μ if $\mu(E) = 0$.

Definition 4.33 (Complete). μ is a complete measure if $F \subseteq E \in M$ and $\mu(E) = 0$ implies $F \in M$.

Proposition 4.34. Given a measure space (X, M, μ) , we can let

$$\overline{M} = \{E \cup F : E \in M, \text{ there exists } G \supset F \text{ with } \mu(G) = 0\}.$$

$\overline{M}$ is a σ -algebra and there is a unique extension $\bar{\mu}$ of μ to a complete measure on $\overline{M}$.

Proof. To show $\overline{M}$ is a σ -algebra, we need to check closure under complements and countable unions.

Pick $E \cup F \in \overline{M}$ using the above notation, i.e. F is contained in some null set $G \in M$. $E \in M$ too so $E^c \cap G^c \in M$. Hence $(E \cup F)^c = E^c \cap F^c = (E^c \cap G^c) \cup (E^c \cap F^c \cap G) \in \overline{M}$.

Say $(E_i \cup F_i) \subseteq \overline{M}$ and $F_i \subseteq G_i$ for null-sets $G_i \in M$. Thus, $\cup (E_i \cup F_i) = \cup (E_i) \cup \cup (F_i)$. $\cup E_i \in M$ and $\cup F_i \subseteq \cup G_i$ and $\mu(\cup G_i) \leq \sum \mu(G_i) = 0$ so $\overline{M}$ is indeed a σ -algebra.

Define $\bar{\mu}(E \cup F) = \mu(E)$. Clearly, $\bar{\mu}$ will be complete if it is a measure. Say $(E_i \cup F_i) \subseteq \bar{M}$ are disjoint. Then, $\bar{\mu}(\cup(E_i \cup F_i)) = \bar{\mu}(\cup(E_i) \cup \cup(F_i)) = \mu(\cup E_i) = \sum \mu(E_i) = \sum \bar{\mu}(E_i \cup F_i)$.

Such a measure extending μ is unique. That is, by (a) in the above proposition,

$$\mu(E) = \bar{\mu}(E) \leq \bar{\mu}(E \cup F) \leq \bar{\mu}(E \cup G) = \mu(E \cup G) \leq \mu(E) + \mu(G) = \mu(E) \implies \bar{\mu}(E \cup F) = \mu(E).$$

□

We would like to extend our measures to any subset of $\mathcal{P}(X)$.

Definition 4.35 (Outer measure). An **outer measure** on a non-empty subset X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ satisfying $\mu^*(\emptyset) = 0$, $\mu^*(A) \leq \mu^*(B)$ whenever $A \subseteq B$, and $\mu^*(\cup_1^\infty A_i) \leq \sum_1^\infty \mu^*(A_i)$.

Proposition 4.36. Given a function μ with values in $[0, \infty]$ on domain $M \subseteq \mathcal{P}(X)$ containing $\emptyset, X$ such that $\mu(\emptyset) = 0$, define

$$\mu^*(A) = \inf \left\{ \sum_1^\infty \mu(E_i) : A \subseteq \cup_1^\infty E_i, E_i \in M \right\}.$$

μ^* is an outer measure.

Proof. First of all, $0 \leq \mu^*(\emptyset) \leq \sum_1^\infty \mu(\emptyset) = 0$. Next, if $A \subseteq B$, then any cover by elements of M for B will also cover A so $\mu^*(A) \leq \mu^*(B)$. Lastly, for any $\epsilon > 0$, we can find $E_{i,j} \in M$ so $A_i \subseteq \cup_{j=1}^\infty E_{i,j}$ and $\sum_{j=1}^\infty \mu(E_{i,j}) - \mu^*(A_i) < \epsilon/2^i$. Thus, $\mu^*(A) \leq \sum_{i,j=1}^\infty \mu(E_{i,j}) < \sum_1^\infty (\mu^*(A_i) + \sum \epsilon/2^i) = \sum_{i=1}^\infty \mu^*(A_i) + \epsilon$. Since ϵ is arbitrary, $\mu^*(A) \leq \sum_1^\infty \mu^*(A_i)$. □

Definition 4.37 (μ^* -measurable). If μ^* is an outer measure on X , a set A is **μ^* -measurable** if $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$ for all $E \subseteq X$.

In particular, A is μ^* -measurable if and only if $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ for any E of finite μ^* -measure.

Theorem 4.38 (Carathéodory). If μ^* is an outer measure on X , the μ^* -measurable sets form a σ -algebra M and μ^* restricted to M is a complete measure.

Proof. Clearly, M is nonempty since for any E , $\mu^*(E) \geq \mu^*(E \cap X) + \mu^*(E \cap X^c) = \mu^*(E) + 0$. M is trivially closed under complements. If $A, B \in M$, then

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \geq [\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c)] + [\mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)].$$

But this sum is greater than or equal to $\mu^*(E)$ by definition of outer measure so $A \cup B \in M$. Thus, if $(A_i) \subseteq M$ are disjoint (which suffices since we have complements), then for any E of finite μ^* -measure,

$$\begin{aligned} \mu^*(E) &\leq \mu^*(E \cap \bigcup_1^\infty A_i) + \mu^*(E \cap (\bigcup_1^\infty A_i)^c) = \mu^*(\bigcup_1^\infty (E \cap A_i)) + \mu^*(E \cap (\bigcup_1^\infty A_i)^c) \\ &\leq \lim_{n \rightarrow \infty} \sum_1^n \mu^*(E \cap A_i) + \mu^*(E \cap (\bigcup_1^n A_i)^c) = \lim_{n \rightarrow \infty} \mu^*(E \cap \bigcup_1^n A_i) + \mu^*(E \cap (\bigcup_1^n A_i)^c) = \mu^*(E). \end{aligned}$$

Now, we show $\mu^*|_M$ is a complete measure. If $F \in M$ and F is a subset of a null-set E in M , then for any finite μ^* -measure G ,

$$\mu^*(G \cap F) + \mu^*(G \cap F^c) \leq 0 + \mu^*(G \cap F^c) \leq \mu^*(G) \implies F \in M.$$

□

Definition 4.39 (Simple μ -measurable function). Say (X, M, μ) is a complete measure space and B/k is a normed vector space. A **simple measurable function** is any $f : X \rightarrow B$ of the form $f = \sum_1^n y_i 1_{E_i}$ for distinct non-zero $y_i \in B$ and distinct non-empty $E_i \in M$.

Definition 4.40 (Simple μ -integrable function). A simple μ -measurable function $f = \sum_1^n y_i 1_{E_i}$ is simple μ -integrable if $\mu(E_i) < \infty$ for each $1 \leq i \leq n$.

Remark 4.41. Simple μ -measurable (integrable) functions form a k -algebra. Plus, if $f : X \rightarrow B$ is a simple μ -measurable (integrable) function, then $\|f\| : X \rightarrow \mathbb{R}$ is as well.

Definition 4.42 (Simple Integral). The integral of a simple μ -integrable function $f = \sum_1^n y_i 1_{E_i}$ is

$$\int_X f d\mu = \sum_1^n y_i \mu(E_i).$$

Definition 4.43 (Measurable function). A function $f : (X, M, \mu) \rightarrow (B/k, \|\cdot\|)$ is μ -measurable if there is a sequence of simple μ -measurable functions f_n which converges to f pointwise (a.e.).

Remark 4.44. Measurable functions form a k -algebra.

We omit the proof of this alternative definition of measurable functions.

Theorem 4.45. A function $f : (X, M, \mu) \rightarrow (B, \|\cdot\|)$ is μ -measurable if and only if $\text{im} f$ is separable and $f^{-1}(U - 0) \in M$ for any open $U \subseteq B$.

Corollary 4.46. If f_n are μ -measurable and converge to f pointwise a.e., then f is also μ -measurable. If f is μ -measurable, so are $f 1_E$ ($E \in M$) and $\|f\|$.

Lemma 4.47. For a simple μ -integrable function f ,

$$\|\int_X f d\mu\| = \|\sum y_i \mu(E_i)\| \leq \sum \|y_i\| \mu(E_i) = \int_X \|f\|.$$

Definition 4.48 (Convergent/Cauchy in measure). A sequence of μ -measurable functions (f_n) are convergent in measure to some μ -measurable f if for all $\epsilon > 0$:

$$\lim \mu\{x \in X : \|f(x) - f_n(x)\| \geq \epsilon\} = 0.$$

A sequence of μ -measurable functions (f_n) are Cauchy in measure if for all $\epsilon > 0$:

$$\lim \mu\{x \in X : \|f_n(x) - f_m(x)\| \geq \epsilon\} = 0.$$

Remark 4.49. (f_n) converges in measure to f, g if and only if $f = g$ almost everywhere.

Definition 4.50 (Mean-Cauchy). A sequence of simple μ -integrable functions (f_n) is mean-Cauchy if $\lim_{n,m \rightarrow 0} \|\int_X (f_n - f_m) d\mu\| = 0$.

Definition 4.51 (Integrable function). A function $f : (X, M, \mu) \rightarrow (B, \|\cdot\|)$ is μ -integrable if there is a mean-Cauchy sequence of simple μ -integrable functions (f_n) that converge to f in measure. All such functions make up the space $\mathcal{L}^1(X, M, \mu, B)$.

To allow for limits of integrals, we need to switch B to a Banach space.

Definition 4.52 (Integral). If (X, M, μ) is a complete measure space and B is a Banach space, consider an integrable function f with a mean-Cauchy sequence of simple μ -integrable functions f_n for which $f_n \rightarrow f$ in measure. The integral of f is

$$\|f\|_1 := \int_X f d\mu := \lim \int_X f_n d\mu.$$

Remark 4.53. Since $\|\cdot\| : B \rightarrow \mathbb{R}$ is continuous, our last lemma implies $\|f\|_1 \leq \int_X \|f\| d\mu$. In particular, $\|f\|_1 \geq \int_X 0 d\mu = 0$. It follows easily from other checks that $\|\cdot\|_1$ is a semi-norm on $\mathcal{L}^1(X, M, \mu, B)$.

Lemma 4.54. $f = g$ almost everywhere if and only if $\|f - g\|_1 = 0$ for $f, g \in \mathcal{L}^1(X, M, \mu, B)$.

$L^1(X, M, \mu, B) = \mathcal{L}^1(X, M, \mu, B) / \{f : \|f\|_1 = 0\}$ is a Banach space. In fact, L^1 is the completion of simple μ -integrable functions.

4.7. Riesz-Markov-Kakutani Theorem

Definition 4.55 (Borel measure). A Borel measure is a measure on the Borel algebra of a topological space X .

Definition 4.56 (Radon measure). A Borel measure μ on X is a Radon measure if:

- (a) μ is locally finite – for any $x \in X$, there is a neighborhood U of x with $\mu(U) < \infty$.
- (b) μ is outer regular – for any Borel set E , $\mu(E) = \inf\{\mu(U) : E \subseteq U \text{ open}\}$.
- (c) μ is inner regular – for any open set U , $\mu(U) = \sup\{\mu(K) : \text{compact } K \subseteq U\}$.

From now on, X will be a locally compact Hausdorff space.

Remark 4.57. For an open subset $U \subseteq X$, we write $f \prec U$ if $\text{supp} f \subseteq U$ and $f \subseteq [0, 1]$.

Definition 4.58 (T_μ). Given any Radon measure μ , we can define a linear functional T_μ on $C_c(X, \mathbb{R})$ via $T_\mu f = \int_X f d\mu$.

In particular, $\text{supp} f$ is compact so, because Radon measures are locally finite, has finite measure. Thus, $T_\mu : C_c(X) \rightarrow \mathbb{R}$ is a linear functional, and obviously positive.

In fact, there is a converse to this statement.

Theorem 4.59 (Riesz-Markov-Kakutani). If X is a locally compact Hausdorff space and T is a positive linear functional on $C_c(X, \mathbb{R})$, then there is a unique Radon measure μ on X for which $Tf = \int_X f d\mu$.

First, we show uniqueness of such a Radon measure μ .

Proof. (Uniqueness). By the outer regularity of Radon measures, it suffices to determine μ on open sets. Fix an open set U and compact $K \subseteq U$. 4.18 promises some $f \prec U$ with $f|_K = 1$ so:

$$\mu(K) = \int_K d\mu \leq \int f d\mu = Tf = \int f d\mu \leq \int_U d\mu = \mu(U) \stackrel{\text{inn. reg.}}{=} \sup_{\text{cpt } K \subseteq U} \mu(K).$$

Thus,

$$\mu(U) = \sup_{\text{cpt } K \subseteq U} \mu(K) = \sup_{f \prec U, f \in C_c(X)} Tf \text{ is determined by } T.$$

□

And now existence.

Proof. (Existence). First, define μ on the open sets of X via:

$$\mu(U) := \sup\{Tf : f \prec U, f \in C_c(X)\}.$$

Extend μ^* to $\mathcal{P}(X)$ via $\mu^*(E) = \inf\{\mu(U) : E \subseteq U \text{ open}\}$. We show μ^* is an outer measure. Of course, if $A \subseteq B \subseteq U$, then $\mu^*(A) \leq \mu^*(B)$. If $(A_i) \subseteq \mathcal{P}(X)$, then $\mu^*(\bigcup_1^\infty A_i) \leq \sum \mu^*(A_i)$ as long as $\mu(\bigcup_1^\infty U_i) \leq \sum_1^\infty \mu(U_i)$ for open sets $U_i \subseteq X$. Fix $f \prec \bigcup_1^\infty U_i$ with $f \in C_c(X)$.

Since $\text{supp} f$ is compact, $f \prec \bigcup_1^n U_i$ without loss of generality. By 4.19, we get a partition of unity $f_i \prec U_i$ with $f_i \in C_c(X)$ so $\sum f_i \equiv 1$ on $\text{supp} f$. By the positivity of T in the first inequality,

$$Tf \leq T\left(\sum_1^n f_i\right) = \sum_1^n Tf_i \leq \sum_1^n \mu(U_i) \leq \sum_1^\infty \mu(U_i).$$

Taking the supremum over such f finishes so μ^* is an outer measure. Now, 4.38 shows μ^* restricts to a complete measure on the μ^* -measurable sets.

In particular, fix an open set U and say $E \subseteq X$ has finite μ^* -measure. For any $\epsilon > 0$, there is some open set V containing E for which $\mu^*(V) \leq \mu^*(E) + \epsilon$. There is $f \prec V \cap U$ in $C_c(X)$ with $\mu(V \cap U) < Tf + \epsilon$. $V \cap U^c \subseteq V \cap \text{supp} f^c$ is open so there is $g \prec V \cap \text{supp} f^c$ with $\mu(V \cap \text{supp} f^c) < Tg + \epsilon$. In particular, $f + g \prec V$ and $f + g \in C_c(X)$ so

$$\mu^*(E \cap U) + \mu^*(E \cap U^c) \leq \mu^*(V \cap U) + \mu^*(V \cap U^c) \leq T(f + g) + 2\epsilon \leq \mu(V) + 2\epsilon < \mu^*(E) + \epsilon.$$

Taking $\epsilon \rightarrow 0$ shows all open sets are μ^* -measurable, and thus μ^* restricts to Borel measure. Next we show μ^* is locally finite.

Fix a compact subset $K \subseteq X$. Say $f \in C_c(X)$, $f \geq \chi_K$, and fix some $\epsilon > 0$. Let $U = \{x \in X : f(x) > 1 - \epsilon\}$. If $g \prec U$ and $g \in C_c(X)$, then $g(x) \leq (1 - \epsilon)^{-1}f$ so $Tg \leq (1 - \epsilon)^{-1}Tf$. Thus $\mu(K) \leq \mu(U) \leq (1 - \epsilon)^{-1}Tf \xrightarrow{\epsilon \rightarrow 0} Tf$ so $\mu(K) \leq \inf_{f \geq \chi_K, f \in C_c(X)} Tf$. On the other hand, there is some open neighborhood U of K for which $\mu(U) \leq \mu^*(K) + \epsilon$. By Urysohn's lemma, there is some $f \in C_c(X)$ with $f \geq \chi_K$ and $f \prec U$ such that $Tf \leq \mu(U) \leq \mu^*(K) + \epsilon$. Taking $\epsilon \rightarrow 0$ and the infimum shows

$$\mu^*(K) = \inf\{Tf : f \geq \chi_K, f \in C_c(X)\}.$$

In particular, if K is compact, then, by Urysohn's lemma, there is a $g \in C_c(X)$ with $g \leq [0, 1]$ so $g|_K = 1$. So $\mu^*(K) \leq Tf < \infty$ since T maps into $\mathbb{R}$ and is positive. Points are compact so μ^* is locally finite.

μ^* is outer regular by definition so we are left to show μ^* is inner regular. Fix an open set U . Pick $f \in C_c(X)$ with $f \prec U$. By the positivity of T , $Tf \leq Tg$ so

$$Tf \leq \inf\{Tg : g \in C_c(X), \chi_{\text{supp} f} \leq g\} = \mu^*(\text{supp} f) \leq \sup_{\text{cpt } K \subseteq U} \mu^*(K).$$

Taking the supremum over f shows $\mu(U) \leq \sup_{\text{cpt } K \subseteq U} \mu^*(K)$. On the other hand, if $f \geq \chi_K$ for some compact subset K , then Urysohn's lemma implies we have some $g \in C_c(X)$ with $g \geq \chi_K$ and $g \prec U$, i.e.

$$\mu^*(K) = \inf\{Tf : f \geq \chi_K, f \in C_c(X)\} \leq Tg \leq \sup\{Tf : f \in C_c(X), f \prec U\} = \mu(U).$$

Taking the supremum over compact subsets of U finishes.

Lastly, we show that $Tf = \int_X f d\mu^*$ for any $f \in C_c(X)$. Let $K_0 = \text{supp} f$ and $K_n = \{x \in X : f \geq n/N\}$ for $1 \leq n \leq N$. K_0 is compact by assumption and since K_n is a closed subset of K_0 , it is also compact. Put

$$f_n(x) = \begin{cases} \frac{1}{N} & x \in K_n, \\ f(x) - \frac{n-1}{N} & x \in K_{n-1} - K_n, \\ 0 & x \notin K_{n-1}. \end{cases}$$

$\text{supp} f_n \subseteq K_{n-1}$ so $f_n \in C_c(X)$. Hence,

$$\frac{\chi_{K_n}}{N} \leq f_n \leq \frac{\chi_{K_{n-1}}}{N} \implies \frac{\mu^*(K_n)}{N} \leq \int_X f_n d\mu \leq \frac{\mu^*(K_{n-1})}{N}.$$

Since $\mu^*(K_n) = \inf\{Tf : f \in C_c(X), f \geq \chi_{K_n}\}$, $\mu^*(K_n) \leq N \cdot Tf_n$. If U is an open set containing K_{n-1} , $Nf_n \prec U$ so $Tf_n \leq \mu^*(U)/N$. Since $\sum_1^N f_i = f$, $\frac{1}{N} \sum_1^N \mu^*(K_n) \leq Tf$, $\int_X f d\mu \leq \frac{1}{N} \sum_1^N \mu^*(K_{n-1})$

$$\left| Tf - \int_X f d\mu \right| \leq \left| \frac{\mu^*(K_{n-1}) - \mu^*(K_0)}{N} \right| \leq \mu^*(K_0)/N.$$

We already showed that $\mu^*(K_0) < \infty$ since K_0 is compact, taking $N \rightarrow \infty$ shows

$$Tf = \int_X f d\mu.$$

□

§5. Functional Analysis

5.1. C*-algebras

Definition 5.1 (Banach algebra). A **Banach algebra** $(B, \|\cdot\|)$ is an associative algebra over $\mathbb{C}$ that is also a complete normed space for which

$$\|ab\| \leq \|a\| \|b\|.$$

Remark 5.2. If a Banach algebra B does not have a unit, we can consider $B^+ = B \times \mathbb{C}$ with unit $(0, 1)$ and respective algebra multiplication and norm

$$(a, \lambda) * (b, \mu) := (ab + a\mu + b\lambda, \lambda\mu), \quad \|(a, \lambda)\| := \sup_{b \in B} \{\|ab + \lambda b\| : \|b\| \leq 1\}.$$

Definition 5.3 (C*-algebra). A **C*-algebra** is a Banach algebra $(A, \|\cdot\|)$ with an involution (antilinear isometry) $a \mapsto a^*$ satisfying:

$$(\lambda a + \mu b)^* = \bar{\lambda} a^* + \bar{\mu} b^*, \quad (ab)^* = b^* a^*, \quad a^{**} = a, \quad \|a^* a\| = \|a\|^2.$$

Remark 5.4. If $\|a^2\| = \|a^* a\| \leq \|a^*\| \|a\|$, then $\|a\| \leq \|a^*\|$. We have the symmetric inequality so $\|a\| = \|a^*\|$.

Remark 5.5. If A is a C*-algebra, so is A^+ .

Remark 5.6. If J is a closed (two-sided) ideal in a Banach algebra A , then A/J is also a Banach algebra under the quotient norm $\|a + J\| := \inf_{j \in J} \|a + j\|$. A/J will also be unital (C*) if A is unital (C*).

Lemma 5.7. The set of invertible elements $A^\times$ of a unital Banach algebra A is open.

Proof. Fix an invertible element x . Pick y with $\|x - y\| < \|x^{-1}\|^{-1}$ so $\|1 - x^{-1}y\| \leq \|x^{-1}\| \|x - y\| < 1$. It follows that $\sum_{k=0}^{\infty} (1 - x^{-1}y)^k$ has norm upper-bounded by $(1 - \|1 - x^{-1}y\|)^{-1}$ so will converge to $(x^{-1}y)^{-1} = y^{-1}x$. Multiplying by x^{-1} isolates y^{-1} so y^{-1} is invertible. $\square$

Corollary 5.8. No proper ideal in a unital Banach algebra is dense.

Definition 5.9 (Resolvent). The **resolvent** of an element a in a unital C*-algebra A is

$$R_a(\lambda) = (\lambda 1 - a)^{-1}.$$

R_a is defined (and holomorphic) on $R_a^{-1} A^\times$, an open subset of $\mathbb{C}$.

Remark 5.10. Notice $(\lambda 1 - a)^{-1} = \lambda^{-1} \sum_{k=0}^{\infty} (\lambda^{-1} a)^k$ converges for $\|\lambda^{-1} a\| < 1$, i.e. for $\|a\| < |\lambda|$. $R_a \rightarrow 0$ as $\lambda \rightarrow \infty$ so if R_a could be extended to all of $\mathbb{C}$, then Liouville's theorem would force R_a to vanish everywhere, which makes no sense since $R_a(\lambda)$ is invertible.

It follows that the subset on which R_a is undefined is both non-empty and a closed subset of the compact $\|a\|$ -ball in $\mathbb{C}$, hence compact.

Definition 5.11 (Spectrum). The **spectrum** of some $a \in A$ is the non-empty, compact subset

$$\sigma(a) := \{\lambda \in \mathbb{C} : \lambda 1 - a \text{ not invertible}\} \subseteq \overline{B_{\|a\|}(0)}.$$

Proposition 5.12. The smallest disc centered at 0 containing $\sigma(a)$ has radius $r(a) := \lim \|a^n\|^{1/n}$.

Proof. Since $\lambda^n - a^n = \lambda^n(1 - \lambda^{-n}a^n) = \lambda^n(1 - \lambda^{-1}a)(1 + \lambda^{-1}a + \dots + \lambda^{-n+1}a^{n-1})$. Thus, if $\lambda^n \notin \sigma(a^n)$, then $\lambda \notin \sigma(a)$. In particular, $\lambda^n \leq \|a^n\|$ so taking the supremum over $\lambda \in \sigma(a)$ gives $r(a) \leq \lim \|a^n\|^{1/n}$.

For the other direction, $R_a(\lambda) = \sum_{k=0}^{\infty} \lambda^{-k-1} a^k$ converges on $|\lambda| \geq B_{r(a)}(0)$. Hence, $R_a(\lambda^{-1})$ converges on $B_{r(a)^{-1}}(0)$ so $r(a)^{-1} \leq (\limsup \lambda^{(k+1)/k} \|a^k\|^{1/k})^{-1}$ – the power series radius of convergence. In particular $r(a) \geq \lim \|a^n\|^{1/n}$ so we are done. $\square$

Corollary 5.13. If a, b commute then

$$r(ab) = \lim \|(ab)^n\|^{1/n} = \lim \|a^n b^n\|^{1/n} \leq \lim \|a^n\|^{1/n} \|b\|^{1/n} = r(a)r(b).$$

Remark 5.14. By 5.16, $\lambda \in \sigma(a)$ if and only if $\lambda - a$ is not invertible if and only if $\bar{\lambda} - a^*$ is not invertible so $\sigma(a)$ is the conjugate of $\sigma(a^*)$. Hence $r(a) = r(a^*)$

Remark 5.15. If $a \in A$ and A is a non-unital Banach algebra, we define the spectrum of a to be its spectrum of $(a, 0) \in A^+$. Elements of A are not invertible in A^+ so 0 is in the spectrum of any $a \in A$ automatically.

From now on, we take A to be a C^* -algebra.

Remark 5.16. If A is unital, then $1 = 1^*$ since for any $a \in A$,

$$a = a^{**} = (a^* \cdot 1)^* = 1^* \cdot a.$$

And $a \in A^\times$ if and only if $a^* \in A^\times$ since

$$1 = (a \cdot a^{-1})^* = (a^{-1})^* \cdot a^*.$$

It follows that $\|1\| = 1$ by the C^* -algebra condition:

$$\|1\| = \|11^*\| = \|1\|^2.$$

Lemma 5.17. Say u is invertible (i.e. $0 \notin \sigma(u)$). Then $\sigma(u^{-1}) = \{\lambda^{-1} : \lambda \in \sigma(u)\}$.

Proof. Notice $\lambda - u = \lambda u(u^{-1} - \lambda^{-1})$. □

Definition 5.18 (Unitary). An element $u \in A$ is unitary if $uu^* = u^*u = 1$.

Lemma 5.19. Unitary elements have spectra in S^1 .

Proof. Say u is unitary. Thus $\|u\|^2 = \|uu^*\| = \|1\| = 1$ so $\|u\| = 1$, implying $\sigma(u), \sigma(u^*) \subseteq B_1(0)$. But u is invertible so 5.17 applies and $\sigma(u) \subseteq S^1$. □

Definition 5.20 (Self-adjoint, normal). $a \in A$ is self-adjoint if $a = a^*$ and normal if $aa^* = a^*a$.

Remark 5.21. If a is self-adjoint, then $\|a^2\| = \|aa^*\| = \|a\|^2$ so $r(a) = \lim \|a\|$. Notice that aa^* is self-adjoint for all $a \in A$. Therefore, if a is normal then, per 5.13 and 5.4,

$$\|a\|^2 = \|aa^*\| = r(aa^*) \leq r(a)r(a^*) \leq \|a\| \|a^*\| = \|a\|^2 \quad r(a) \stackrel{r(a)=r(a^*)}{=} r(a) = \|a\|.$$

Proposition 5.22. Self-adjoint elements have real spectra.

Proof. Suppose $a \in A$ is self-adjoint. Define

$$\exp(ia) := \sum_{k=0}^{\infty} (ia)^k / k!,$$

which converges in A since it is upper-bounded by $e^{\|a\|}$. Observe that

$$\exp(ia)^* = \sum_{k=0}^{\infty} (-ia^*)^k / k! \stackrel{a=a^*}{=} \exp(-ia) = \exp(ia)^{-1}.$$

$\exp(ia)$ is unitary so $\sigma(\exp(ia)) \subseteq S^1$ by 5.19. It follows that $\lambda \in \mathbb{R}$ if and only if $e^{i\lambda} - e^{ia} = e^{i\lambda}(1 - e^{i(a-\lambda)}) = e^{i\lambda}(a - \lambda)(\sum_{n=1}^{\infty} i^n (a - \lambda)^{n-1} / n!)$ is not invertible only if $a - \lambda$ is not invertible, i.e. $\sigma(a) \subseteq \mathbb{R}$. □

Remark 5.23. A morphism of C^* -algebras is a $*$ -homomorphism. A map $f : A \rightarrow B$ extends uniquely to a unital map $f^+ : A^+ \rightarrow B^+$.

Lemma 5.24. A map $f : A \rightarrow B$ of C^* -algebras is norm-decreasing and thus continuous.

Proof. We may assume f is a map on unital C^* -algebras so $\lambda 1_B - f(a) = f(\lambda 1_A - a)$. Hence $\lambda \in \sigma(f(a))$ implies $\lambda \in \sigma(a)$ so $\sigma(f(a)) \subseteq \sigma(a)$. In particular, $r(f(a)) \leq r(a)$ so:

$$\|f(a)\|^2 = \|f(a)f(a)^*\| = \|f(aa^*)\| \stackrel{5.21}{=} r(aa^*) \leq r(a)r(a^*) \leq \|a\| \|a^*\| = \|a\|^2.$$

Hence, f is norm-decreasing. $\square$

5.2. Commutative Gelfand-Naimark Theorem

Definition 5.25 (Character). A character of a Banach algebra A is a nonzero (thus surjective) homomorphism $A \rightarrow \mathbb{C}$. The collection of characters is written $M(A)$.

Remark 5.26. Characters on A necessarily extend to A^+ via $\mu^+(0, 1) = 1$. Notice the zero functional on A thus extends to a character on A^+ so $M(A^+) \longleftrightarrow M(A) \cup \{0\}$.

Remark 5.27. If $\mu(a) \notin \sigma(a)$, then $\mu(a)1_A - a$ is invertible yet its image under μ is 0 - contradiction. So $\mu(a) \in \sigma(a)$ for $\mu \in M(A)$.

It follows that $|\mu(a)| \leq \|a\|$ so $\|\mu\| \leq 1$. $\mu \in M(A)$ is nonzero so $\mu(a) = \mu(a)\mu(1)$ for all $a \in A$ so $\mu(1) = 1$. Hence, $\|\mu\| = 1$.

Corollary 5.28. If a is a self-adjoint element of a C^* -algebra A , then $\mu(a) \in \mathbb{R}$ for all $\mu \in M(A)$, per 5.22.

Definition 5.29 (Gelfand spectrum). If A is commutative, $M(A)$ is the **Gelfand spectrum of A** under the **Gelfand topology** is given by $M(A) \hookrightarrow B_1(0) \subseteq A^\vee$, which is compact under the weak-* topology per Banach-Alaoglu theorem.

Lemma 5.30. The Gelfand spectrum of a commutative Banach algebra A is locally compact.

Proof. We show $M(A) \cup \{0\}$ is closed under weak-* so compact. In particular, let $\phi_{ab}(\mu) := \mu(a)\mu(b) - \mu(ab)$ is a continuous linear functional on $A^\vee$ for fixed $a, b \in A$. By definition, $M(A) \cup \{0\} = \int_{a,b \in A} \phi_{ab}^{-1}(0)$ so is weak-* closed. And $M(A) = [M(A) \cup \{0\}] \cap (\mu \mapsto \mu(1))^{-1}(1)$ $\square$

Definition 5.31 (Gelfand transform). If A is a commutative Banach algebra, the **Gelfand transform** of $a \in A$ is the (continuous) evaluation map $\hat{a} : M(A) \rightarrow \mathbb{C}$ at a .

In particular, $A \rightarrow C_0(M(A))$ via $a \mapsto \hat{a}$ is the Gelfand transform.

Remark 5.32. For any $a \in A$, write $a = \frac{1}{2}(a^* + a) + \frac{i}{2}(ia^* - ia) = a_1 + ia_2$ for a_1, a_2 self-adjoint so will be real-valued under μ . Obviously, then, $\mu(a^*) = \mu(a_1) - i\mu(a_2) = \overline{\mu(a)}$. In other words, $\hat{a}^* = \hat{a}^*$.

Theorem 5.33 ((Complex) Gelfand-Mazur). A unital Banach algebra $A/\mathbb{C}$ that is also a field is isometrically isomorphic to $\mathbb{C}$.

Proof. Fix $a \in A$. $\sigma(a)$ contains some λ_a by 5.10. $\lambda_a 1_A - a$ is non-invertible in A , a field, so $a - \lambda_a 1_A = 0$. It's obvious that $a \mapsto \lambda_a$ is a nonzero injective ring homomorphism into $\mathbb{C}$. Since A is a $\mathbb{C}$ -vector space, $A \cong \mathbb{C}$. $\|a\| = \|\lambda_a 1_A\| = |\lambda_a| \|1_A\|$ so this is obviously an isometry. $\square$

Remark 5.34. The above isomorphism is canonical!

Lemma 5.35. Say A is a commutative C^* -algebra and $\lambda \in \sigma(a)$ for some $a \in A$. If A is unital (or if A is non-unital and $\lambda \neq 0$), there is a character μ so that $\mu(a) = \lambda$.

Proof. Suppose A is unital. For $\mu \in M(A)$, $A/\ker \mu \cong \mathbb{C}$ implying $\ker \mu$ is a maximal ideal.

On the other hand, say I is a maximal ideal. Since ring operations are continuous, its closure is also an ideal. Namely, $\bar{I}$ is maximal since I cannot be dense by 5.8. Thus, $I = \bar{I}$ so its quotient map $\eta : A \twoheadrightarrow A/I$ produces a unital commutative Banach algebra and field A/I , which is canonically isomorphic to $\mathbb{C}$ by 5.33. Thus, $\tilde{\eta} : A \rightarrow \mathbb{C}$ so $\tilde{\eta} \in M(A)$.

By Zorn's lemma, $A(\lambda 1_A - a)$ is contained in some maximal ideal I (nontrivial since $\lambda \in \sigma(a)$ by assumption) for which $\tilde{\eta}(I) = 0$, i.e. $\tilde{\eta}(a) = \lambda \tilde{\eta}(1) = \lambda$.
If A is non-unital, then doing the proof for A^+ suffices and we can restrict the promised character back to A . $\square$

Corollary 5.36. Suppose A is a commutative C^* -algebra and fix $a \in A$. For any $\lambda \in \sigma(a)$, there is some $\mu \in M(A)$ (where we've shown $\|\mu\| \leq 1$) with $\hat{a}(\mu) = \mu(a) = \lambda$, i.e. $\|\hat{a}\| = r(a)$.

We are finally ready to conclude the following.

Theorem 5.37 (Gelfand-Naimark). If A is a commutative C^* -algebra, the Gelfand transform is an isometric $*$ -isomorphism of A onto $C_0(M(A))$.

Proof. The Gelfand transform $a \mapsto \hat{a}$ is a $*$ -homomorphism because of 5.32 and the fact that the evaluation map is a ring homomorphism. The transform is an isometry since:

$$\|\hat{a}\|^2 = \|\hat{a}\hat{a}^*\| = \|\widehat{aa^*}\| \stackrel{5.36}{=} r(aa^*) = \|a^*a\| = \|a\|^2.$$

It follows that the transform is injective.

Hence, A is a subalgebra of $C_0(M(A))$. If $\mu, \eta \in M(A)$ differ, then they must differ on some $a \in A$, so $\hat{a}$ separates them. Also, since characters are nonzero homomorphisms, they cannot vanish on all of A , so A vanishes nowhere on $M(A)$. Thus, Stone-Weierstraß (4.24) implies that the Gelfand transform will surject onto $C_0(M(A))$ as long as A is closed in $C_0(M(A))$. But, this follows because it is a complete subalgebra. $\square$

5.3. Continuous functional calculus

Remark 5.38. Let $C^*(a, 1)$ denote the norm-closure of the sub-algebra of a unital C^* -algebra A , generated by $a, a^*, 1$. If a is normal, $C^*(a, 1)$ is commutative.

Lemma 5.39. $\hat{a} : M(C^*(a, 1)) \rightarrow \sigma(a)$ via $\mu \mapsto \mu(a)$ is a homeomorphism.

Proof. $M(C^*(a, 1))$ is compact and $\hat{a}$ is clearly a continuous bijection so a homeomorphism. $\square$

Theorem 5.40 (Continuous functional calculus). If a is normal, then $C(\sigma(a), \mathbb{C}) \rightarrow C^*(a, 1)$ via $f(x) \mapsto f(a)$ is an isometric $*$ -isomorphism.

Proof. Since a is normal, $C^*(a, 1)$ is a commutative C^* -algebra so Gelfand-Naimark theorem (5.37) implies $C^*(a, 1) \cong C_0(M(C^*(a, 1))) \stackrel{5.39}{\cong} C_0(\sigma(a)) = C(\sigma(a))$. $\square$

Corollary 5.41 (Spectral Mapping Theorem). If A is a C^* -algebra and $a \in A$ is normal, then $f \in C_0(\sigma(a))$ implies

$$\sigma(f(a)) = f(\sigma(a)).$$

5.4. Concrete Gelfand-Naimark Theorem

Definition 5.42 (Positive). An element $a \in A$ is positive if it is self-adjoint and $\sigma(a) \subseteq [0, \infty)$.

Remark 5.43. We write $a \geq 0$ if it is positive and $a \geq b$ if $a - b \geq 0$.

Lemma 5.44. Suppose A is a unital C^* -algebra, $a \in A$ is self-adjoint, and $\lambda \geq \|a\|$ for $\lambda \in \mathbb{R}$. Then a is positive if and only if $\|\lambda 1_A - a\| \leq \lambda$.

Proof. $a \in A$ is self-adjoint so $\sigma(a) \subseteq [-\|a\|, \|a\|]$. So a is positive if and only if $\sigma(a) \subseteq [0, \lambda]$ if and only if $\|\lambda 1_{\sigma(a)} - \text{id}_{\sigma(a)}\| \stackrel{5.40}{=} \|\lambda - \text{id}\|_\infty \leq \lambda$. $\square$

Corollary 5.45. If $a, b \geq 0$, then $a + b \geq 0$.

Proof. Put $\lambda = \|a\| + \|b\| \geq \|a + b\|$ so, applying the above lemma to a, b :

$$\|\lambda - (a + b)\| = \|\|a\| + \|b\| - (a + b)\| \leq \|\|a\| - a\| + \|\|b\| - b\| \leq \|a\| + \|b\| = \lambda.$$

Since $(a + b)^* = a^* + b^* = a + b$, the lemma also guarantees this holds only if $a + b \geq 0$. $\square$

Proposition 5.46. An element $a \in A$ is positive only if there is some $b \in A$ for which $a = b^*b$.

Proof. Say $a \in A$ is positive. $\sigma(a) \subseteq [0, \|a\|]$. So we can define $f(z) = \sqrt{z} \in C(\sigma(a))$ which has image $b = f(a)$ in $C^*(a, 1)$ under 5.37. In particular, for real z , $z = \sqrt{z}\sqrt{z}$ so $a = b^*b$. $\square$

There is a converse to this statement, which will come later.

Definition 5.47 (Approximate unit). An approximate unit in a C^* -algebra

$$A$$

is a net $\{u_\alpha\}$ of positive elements of A such that $\|u_\alpha\| \leq 1$ and $\|bu_\alpha - b\| \rightarrow 0$ for each $b \in A$.

Definition 5.48 (Positive). A linear functional $\phi : A \rightarrow \mathbb{C}$ is positive if $\phi(a) \geq 0$ for all $a \geq 0$.

Definition 5.49 (State). A state ϕ is a norm one positive linear functional. It is faithful if $a \geq 0$ and $\phi(a) = 0$ imply $a = 0$.

Definition 5.50 (Trace). A normalized trace is a nontrivial state ϕ with $\phi(ab) = \phi(ba)$ for all $a, b \in A$.

Remark 5.51. The space of states $S(A)$ is a convex set. The extreme points of this convex set are called pure states.

Remark 5.52 (Gelfand-Naimark-Segal Construction). Any state ϕ of a C^* -algebra A induces a representation π_ϕ of A .

The form $(a, b)_\phi := \phi(a^*b)$ is a sesquilinear (linear in one argument and conjugate linear in the other argument) and positive semidefinite ($(a, a) \geq 0$) form on A . By CBS-inequality

$$|\phi(a^*b)|^2 \leq \phi(a^*a)\phi(b^*b).$$

It follows that

$$N_\phi := \{b \in A : \phi(b^*b) = 0\} = \{b \in A : \phi(a^*b) = 0 \text{ on } A\}$$

is a closed left ideal in A . Say H_ϕ is the completion of A/N_ϕ under the induced inner product $(a + N_\phi, b + N_\phi)_\phi$.

Define $\pi_\phi : A \rightarrow \mathcal{L}(H_\phi)$, the bounded operators on H_ϕ , via $a \mapsto (b + N_\phi \mapsto ab + N_\phi)$. This is well defined since

$$\|\pi_\phi(a)(b)\| = \|ab\| = \phi(b^*a^*ab) \leq \|a^*a\|\phi(b^*b) \leq \|a\|^2\|b\|^2 \implies \|\pi_\phi(a)\| \leq \|a\|.$$

It follows that π_ϕ is an algebra homomorphism.

We're ready to mention the second Gelfand-Naimark theorem.

Theorem 5.53 (Gelfand-Naimark). Any C^* -algebra A has an isometric representation as a closed subalgebra of the algebra $\mathcal{L}(H)$ of bounded operators on some Hilbert space H .

Proof. Fix some nonzero positive element $b^*b \in A$. By Hahn-Banach, there is a state ψ with $\psi(b^*b) = \|b^*b\| = \|b\|^2$. Take Φ to be the family of all states $S(A)$. Take the direct sum of GNS representations $\pi = \bigoplus_{\phi \in \Phi} \pi_\phi$ acting on $H = \bigoplus_{\phi \in \Phi} H_\phi$ described in 5.52. Thus $\|\pi(b)\| = \|b\|$ for all $b \in A$ so π is an isometric representation of A on $\mathcal{L}(H)$. $\square$

Corollary 5.54. A self-adjoint element $a \in A$ is positive if and only if $\phi(a) \geq 0$ for all $\phi \in S(A)$.

Corollary 5.55. If η is a unit vector in H , then $(a \mapsto (\eta, \pi(a)\eta)) \in S(A)$ because

$$b^*b \mapsto (\eta, \pi(b^*b)\eta) = (\eta, \pi(b)^*\pi(b)\eta) = \|\pi(b)\eta\|^2 \geq 0.$$

So $\pi(a)$ is a positive operator on H and thus has a square root $\sqrt{\pi(a)}$. Because π is an isomorphism, $\sqrt{\pi(a)} = \pi(b)$ for a unique self-adjoint element $b \in A$. Undoing the isomorphism, $a = b^2$.

Corollary 5.56. If an element can be written as $a = b^*b$, it is positive.

Proof. Clearly, $a = a^*$ so $\sigma(a) \subseteq \mathbb{R}$. We can extend $\phi(a) = \lambda$ for $\lambda \in \sigma(a)$ to all of A with $\|\phi\| = \phi(1)$ making it a state. By CBS-inequality:

$$\lambda = \phi(a) = \phi(b^*b) \geq |\phi(b)|^2 \geq 0 \implies \sigma(a) \subseteq [0, \|a\|].$$

□

5.5. Categorical Perspective

We show the category of compact Hausdorff spaces is equivalent to the opposite category of commutative unital C^* -algebras.

Theorem 5.57. There is a contravariant functor $\mathcal{C}$ from the category of compact Hausdorff spaces with continuous maps to the category of commutative unital C^* -algebras with unital $*$ -homomorphisms.

Proof. Put $\mathcal{C}(X) = C(X, \mathbb{C})$ and $\mathcal{C}(f : X \rightarrow Y) = (Cf : CY \rightarrow CX) = f^*$. Clearly, $\mathcal{C}(\text{id}_X) = \text{id}_{C(X)}$ and $\mathcal{C}(g \circ f) = \mathcal{C}(f) \circ \mathcal{C}(g)$ for $X \xrightarrow{f} Y \xrightarrow{g} Z$. It's obvious that $C(X)$ is a commutative unital C^* -algebra and its morphisms are unital $*$ -algebra homomorphisms. □

In fact, there is a contravariant functor going the other way.

Theorem 5.58. There is a contravariant functor $\mathcal{M}$ from the category of commutative unital C^* -algebras with unital $*$ -homomorphisms to the category of compact Hausdorff spaces with continuous maps.

Proof. Put $\mathcal{M}(A) = M(A)$ the Gelfand spectrum with topology making each $\hat{a} : M(A) \rightarrow \mathbb{C}$ continuous and $\mathcal{M}(\phi : A \rightarrow B) = M(B) \xrightarrow{\phi^*} M(A)$. $\mathcal{M}(\phi)$ is continuous since $\hat{a} \circ \mathcal{M}(\phi) = \widehat{\phi(a)}$ is continuous for all $a \in A$. We've shown that $M(A)$ is compact Hausdorff in 5.30 so this assignment makes sense. It is easy to check that $\mathcal{M}$ is a contravariant functor. □

Lemma 5.59. All characters on $C(X)$ are evaluations on X .

Proof. Any character $\mu \in M(C(X))$ satisfies $\mu : C(X) \rightarrow \mathbb{C}$ so $\ker \mu$ is a maximal ideal of $C(X)$. Suppose no element of X vanishes on $\ker \mu$; thus, for each $x \in X$, there is some $f_x \in \ker \mu$ so $|f_x(x)| > 0$. Thus, $f_x^{-1}(\mathbb{C} - \overline{B}_{|f_x(x)|/2}(0))$ is an open neighborhood of x away from 0. By the compactness of X , we can cover X with finitely many such neighborhoods to produce some $f_1, \dots, f_n$ for which $f = \sum f_i \bar{f}_i > 0$ where $\mu(f) = 0$ since μ is an algebra homomorphism. Yet $\frac{1}{f} \in C(X)$ so $\mu(f)$ is invertible, i.e. non-zero – contradiction. Hence, $\ker \mu$ vanishes at some $x \in X$. Because $\text{ev}_x : C(X) \rightarrow \mathbb{C}$, $\ker \text{ev}_x \supseteq \ker \mu$, both maximal ideals, so they are equal. Applying ev_x to $f - \mu(f)1$ for all $f \in C(X)$ shows $\mu = \text{ev}_x$. □

Corollary 5.60. $X \rightarrow M(C(X))$ via $x \mapsto \text{ev}_x$ is actually a homeomorphism since it is a continuous bijection from a compact space. Namely, it is continuous with respect to the Gelfand topology since that is simply the evaluation of characters on functions in $C(X)$.

Theorem 5.61. $\mathcal{C}$ and $\mathcal{M}$ provide an equivalence of categories.

Proof. For a commutative unital algebra A , $\mathcal{C} \circ \mathcal{M}(A) = C(M(A), \mathbb{C})$ which is naturally isomorphic to the identity functor under the commutative Gelfand-Naimark theorem, 5.37.

Conversely, $\mathcal{M} \circ \mathcal{C}(X) = M(C(X))$ is naturally isomorphic to the identity functor under the homeomorphism given by 5.60. □

Remark 5.62. To generalize this result to locally compact Hausdorff spaces and proper continuous maps with commutative C^* -algebras with $*$ -morphisms, we need to restrict to $*$ -morphisms which send approximate units to approximate units.

Proposition 5.63. Two commutative C^* -algebras are isomorphic if and only if their character spaces are homeomorphic.

Proof. This follows from 5.61 for the unital case. If A, B are non-unital, then they are isomorphic if and only if $A^+ \cong B^+$ if and only if $M(A^+) \cong M(B^+)$ per the categorical equivalence. Recalling from 5.30 that $M(A^+) \longleftrightarrow M(A) \cup \{0\}$ and $\{0\}$ is isolated, we're done. $\square$

§6. Homotopy in C^* -algebras